

RESEARCH ARTICLE

10.1002/2017JE005298

Key Points:

- New systematic experiments reveal behavior of spherical harmonic series near the Moon's topography as function of resolution and altitude
- Gravity from degree-180 models free of divergence, degree-360 models partially divergent, and degree-2160 models severely divergent
- New hypothesis: local minimum in potential degree variances foreshadows series divergence inside the Brillouin sphere

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Citation:

Hirt, C., and M. Kuhn (2017), Convergence and divergence in spherical harmonic series of the gravitational field generated by high-resolution planetary topography—A case study for the Moon, *J. Geophys. Res. Planets*, 122, 1727–1746, doi:10.1002/2017JE005298.

Received 8 MAR 2017

Accepted 21 JUL 2017

Accepted article online 24 JUL 2017

Published online 18 AUG 2017

Convergence and divergence in spherical harmonic series of the gravitational field generated by high-resolution planetary topography—A case study for the Moon

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Abstract Theoretically, spherical harmonic (SH) series expansions of the external gravitational potential are guaranteed to converge outside the Brillouin sphere enclosing all field-generating masses. Inside that sphere, the series may be convergent or may be divergent. The series convergence behavior is a highly unstable quantity that is little studied for high-resolution mass distributions. Here we shed light on the behavior of SH series expansions of the gravitational potential of the Moon. We present a set of systematic numerical experiments where the gravity field generated by the topographic masses is forward-modeled in spherical harmonics and with numerical integration techniques at various heights and different levels of resolution, increasing from harmonic degree 90 to 2160 (~61 to 2.5 km scales). The numerical integration is free from any divergence issues and therefore suitable to reliably assess convergence versus divergence of the SH series. Our experiments provide unprecedented detailed insights into the divergence issue. We show that the SH gravity field of degree-180 topography is convergent anywhere in free space. When the resolution of the topographic mass model is increased to degree 360, divergence starts to affect very high degree gravity signals over regions deep inside the Brillouin sphere. For degree 2160 topography/gravity models, severe divergence (with several 1000 mGal amplitudes) prohibits accurate gravity modeling over most of the topography. As a key result, we formulate a new hypothesis to predict divergence: if the potential degree variances show a minimum, then the SH series expansions diverge somewhere inside the Brillouin sphere and modeling of the internal potential becomes relevant.

Plain Language Summary Planetary scientists often use series expansions to represent gravity field variations. However, this mathematical technique may produce grossly incorrect results when applied with very high resolution on rugged planetary bodies such as the Moon. This is also called series divergence. This study presents new systematic experiments that compare the series expansions with a second mathematical technique called numerical integration. The experiments provide for the first time insights into the behavior of series expansions for modeling of the Moon's gravity field. As main result, the series expansions—as commonly used in planetary sciences—cannot be used to accurately represent km to 10 km details of the Moon's gravity field near the surface of the Moon. The paper identifies the chances and limitations of frequently used modeling techniques for the Moon. It presents a new hypothesis that uses a new, simple indicator for occurrence or absence of series divergence. The methods presented can be applied to other terrestrial planets, e.g., Venus and Mars, to obtain similarly detailed insight into the series expansions.

1. Introduction

In planetary sciences, spherical harmonic (SH) series expansions are commonly used to model the exterior gravitational potential of Moon, Earth, and other terrestrial planets [e.g., Blakeley, 1996; Balmino et al., 2012; Konopliv et al., 2014; Lemoine et al., 2014; Wieczorek, 2015]. The coefficients of the SH series expansions can be either determined from field observations (e.g., measured gravity accelerations or satellite orbit perturbations), cf. Pavlis [2014], or with forward modeling techniques [e.g., Rexer et al., 2016]. In the latter case, a model of the planetary mass distribution (e.g., surface topography model together with mass density assumptions) is used to derive the coefficients of the implied gravitational field [e.g., Rummel et al., 1988; Hirt and Kuhn, 2014].

In either case, the SH series of the gravitational potential or functionals thereof (e.g., radial or horizontal derivatives) converge outside a sphere encompassing all field-generating mass, centered to the origin of the

spherical coordinate system [Hu and Jekeli, 2015]. In the literature, this sphere is known as Brillouin sphere [e.g., Moritz, 1980]. Inside the Brillouin sphere, however, the convergence of the SH series is not guaranteed. They may converge or diverge, i.e., produce invalid numerical values of gravity field functionals [e.g., Jekeli, 1983]. Near or at the surface of the masses, series convergence must be considered an unstable property [Krarup, 1969], whereby “an arbitrarily small change” of the mass distribution may “change convergence to divergence” [Moritz, 1978, p19]. Generally, divergence is thought to occur more likely, the higher the spectral resolution of the gravitational model, the more irregular the planetary body and the deeper the evaluation points are located inside the Brillouin sphere [Wang, 1997; Lowes and Winch, 2012; Hu and Jekeli, 2015].

In recent time, the divergence behavior of low-degree harmonic series expansions of gravity field functionals has been intensively studied for irregularly shaped bodies such as the Martian moons Deimos and Phobos [Hu and Jekeli, 2015]; asteroids, e.g., 433 Eros [Hu, 2012], Castalia, and Bennu [Takahashi et al., 2013; Takahashi and Scheeres, 2014]; and comets, e.g., 67P/Churyumov-Gerasimenko [Reimond and Baur, 2016]. All studies demonstrated substantial divergence for evaluation points inside the Brillouin sphere, occurring already at low spectral resolution, showing the SH series unable to model the near-surface exterior gravity field of irregularly shaped bodies with adequate precision and detail.

Interestingly, the convergence/divergence issue associated with the synthesis of SH gravitational fields of terrestrial planets in general and the Moon in particular has not received much attention recently. Compared to asteroids and other such objects, the Moon and the terrestrial planets are much more regular in shape and reasonably well approximated through rotationally symmetric bodies such as a sphere (e.g., Moon) or ellipsoid (e.g., Earth). As a consequence of the more regular shape compared to asteroids and other oddly shaped bodies, the free space inside the Brillouin sphere is reduced [Hu and Jekeli, 2015]. As a result, the SH gravity models of Moon and planets tend to be less prone to divergence, at least at lower resolution (also see Werner and Scheeres [1997, p314] “in planetary applications, the divergence may not exist or may be ignored as the body will be nearly spherical”). However, the situation is completely different with high-resolution gravity models, as we show in this study for the Moon.

While the gravitational field of the Moon is now routinely modeled with higher degree SH series expansions—particularly as a result of the Gravity Recovery and Interior Laboratory (GRAIL) gravity field mission [Zuber et al., 2013; Lemoine et al., 2014; Konopliv et al., 2014]—surprisingly little is known about if and where inside the Brillouin sphere these series actually converge or diverge. In some cases, series convergence is taken for granted even when the evaluation points are located inside the Brillouin sphere of the Moon [Goossens et al., 2011; Hirt and Featherstone, 2012; Featherstone et al., 2013; Konopliv et al., 2014]. However, divergence in the SH series of gravity field functionals may be a critical factor in high-resolution planetary gravity modeling, as demonstrated in this contribution.

The goal of the present paper is to shed new light on the convergence versus divergence behavior of SH series inside the Brillouin sphere, exemplified here for the external gravitational field of the Moon. We present and analyze a set of systematically designed and conducted numerical experiments where the gravity field of the topographic masses is forward-modeled in SH, and with numerical integration (NI) techniques at different levels of resolution, increasing from degree 90 to 2160. The NI is free from any divergence issues and therefore suitable to reliably assess the convergence versus divergence behavior of the SH models, which are evaluated here at different altitudes (height levels) inside and outside the Brillouin sphere. High-performance computing is deployed to reduce discretization errors in the NI on the one hand. Spectral forward modeling with multiple bandwidths ensures rigorous spectral consistency between the NI and the SH forward gravity model on the other hand.

Because of their widespread application in planetary sciences, the present study focuses on exterior spherical harmonic series, in our case, of gravity values. We acknowledge that some studies discuss spheroidal (ellipsoidal) instead of spherical harmonics to improve the convergence region of the gravity field series expansions [Hu and Jekeli, 2015; Reimond and Baur, 2016]. The ellipsoidal approach, however, does not appear promising for the Moon, given its negligible flattening.

Internal (interior) spherical harmonic series might be an alternative that avoid divergence associated with the widely used exterior series. In local applications, interior harmonics are sometimes used because they guarantee convergence in a sphere exterior and tangent to the planetary surface [Takahashi and Scheeres,

Table 1. Characteristics of the Six Experiments: Resolution of the Topography Model, Oversampling in the Newton Integration, and Maximum Degrees $N_{\max}^G$ and Maximum Powers $p_{\max}$ Modeled in the Spectral Domain

$N_{\max}^T$	Topography Model		Newton Integration		Spectral Gravity Modeling	
	Spatial Resolution (km)	Spectral Resolution (min)	Oversampling Factor of Topography Signal	$N_{\max}^G$ Modeled	$N_{\max}^G$ Required ($\sigma_n^2 > 10^{-25}$)	$p_{\max}$ Modeled (Needed)
90	60.6	120	240	900	400	15 (7)
180	30.3	60	120	1800	900	25 (12)
360	15.2	30	60	3600	3600	50 (40)
720	7.6	15	30	3600	>3600	70 (50)
1080	5.1	10	20	3600	>3600	70 (50)
2160	2.5	5	10	3600	>3600	70 (50)

2014, p171]. However, a combination of interior with exterior harmonics (that is, the two-potential method) can also be a suitable means to obtain gravity values free of divergence, not only locally but also globally. In this approach, widely used for electrostatic field modeling [Jackson, 1962] but not yet so much for gravitational potential fields, there are two series expansions evaluated when the computation point resides inside the Brillouin sphere. The mass below the computation point generates the external potential, and the mass above the computation point produces the internal potential [also see Takahashi *et al.*, 2013, Figure 4 *ibid*]. The correct gravity value is obtained as sum of both potentials, and convergence issues disappear (B. Bills, personal communication, 2017), see also discussion in section 4.

A number of studies discuss or encounter the topic of series convergence versus divergence in the context of Earth's gravity field [e.g., Moritz, 1961, 1978, 1980; Sjöberg, 1980; Jekeli, 1983; Wang, 1997; Shen, 2009; Hirt *et al.*, 2016]. However, most of these focus on gravity modeling with lower resolution than considered here. Also, given the markedly different surface topographies and shape of Earth and Moon, previous results for Earth are not necessarily indicative for the Moon, and this statement is corroborated in the present paper.

2. Data and Methods

2.1. Topographic Mass Models

Topographic mass models are the input data for the gravity forward modeling. They have two components, one is the geometry and the other the density of the mass distribution. In our work, they are constructed based on high-resolution shape maps from the Lunar Orbiter Laser Altimeter (LOLA) instrument [Smith *et al.*, 2010] along with a mass density value derived from GRAIL gravity mission results [Zuber *et al.*, 2013]. We use the degree 2600 MoonTopo2600pa.shape model of the Moon [Wieczorek, 2015] (file available via <http://markwieczorek.github.io/web>) that is an SH expansion of the LOLA 1/64° global grid (file LDEM_64_PA.IMG of March 15, 2013 from the Planetary Data System, <http://pds-geosciences.wustl.edu>). The shape model is given in the principal axis (PA) coordinate system that is consistent with that of GRAIL gravity field models (also see explanations given in the meta-data file LDEM_64_PA.LBL). Planetary radii R_T are computed from shape model's fully normalized SHCs ($\bar{C}_{nm}, \bar{S}_{nm}$) via

$$R_T = \sum_{n=0}^{N_{\max}^T} \sum_{m=0}^n (\bar{C}_{nm} \cos m\lambda + \bar{S}_{nm} \sin m\lambda) \bar{P}_{nm}(\sin \varphi) \quad (1)$$

where $\bar{P}_{nm}$ are the fully normalized Associated Legendre Functions (ALFs) of harmonic degree n and order m , (φ, λ) denote the selenocentric latitude and longitude in the PA system, and $N_{\max}^T$ is the maximum degree in the series expansion. The differences $H = R_T - R$ describe the surface topography of the Moon relative to a reference mass sphere of mean radius R . In all computations, a value of $R = 1,738,000$ m is used, in agreement with the reference radius of the GRAIL gravity field by Lemoine *et al.* [2014] and Konopliv *et al.* [2014].

Our study uses six mass models, with the spectral resolution $N_{\max}^T$ of topographic masses increasing from degree and order 90 (~61 km spatial scales) to 2160 (~2.5 km spatial scales), cf. Table 1. The resolution of the topographic mass models is even multiples of the lowest resolution ($N_{\max}^T = 90$); subsequent resolution levels mostly differ by a factor of 2 (cf. Table 1). The bandwidths of the mass models were chosen such that

divergence issues or numerical problems can be narrowed down as a function of resolution, ranging from ~61 to ~2.5 km.

The surface topography is assigned a uniform mass density value ρ_T of 2500 kg m^{-3} . A mean density value of 2550 kg m^{-3} for the lunar crust was determined by *Wieczorek et al.* [2013] through inversion of GRAIL gravity data. A value of 2450 kg m^{-3} was used by *Konopliv et al.* [2014] in the computation of the topographic reduction of the GRAIL data, while *Lemoine et al.* [2014] assumed a density of 2500 kg m^{-3} in their calculation of Bouguer gravity anomalies. As such, our adopted mass density value ρ_T is the average value used in three recent studies relying on the GRAIL mission and should be a good density approximation for the lunar topography. We note that a constant mass density value will not affect the behavior of the SH series expansions, given the density acts as a "scale factor" (equation (2)).

2.2. Gravity Forward Modeling

2.2.1. General Remarks

In our study, we apply two independent techniques to compute gravitational effects (short: gravity) from topographic mass models (short: topography).

1. The first technique is forward modeling in the spectral domain, whereby the gravitational potential generated by the topography is expanded into spherical harmonic (SH) series of integer powers of the topography [e.g., *Chao and Rubincam*, 1989; *Wieczorek and Phillips*, 1998; *Hirt and Kuhn*, 2012], see section 2.2.3.
2. The second is numerical integration (NI) in the spatial domain, also known as Newtonian integration [e.g., *Werner and Scheeres*, 1997; *Kuhn et al.*, 2009; *Hirt and Kuhn*, 2014], see section 2.2.4.

We apply both techniques with the strategies detailed in *Hirt and Kuhn* [2014] to reach μGal -level consistency for the gravity values when divergence is absent. Both forward modeling techniques use the same topographic mass model in a rigorous, mutually consistent manner. This allows us to study the convergence versus divergence behavior in the evaluation of the SH series expansion of gravity values (cf. equation (4)), in a straightforward way from cross comparisons of forward-modeled gravity from both techniques. While the topography mass models are strictly band-limited to the N_{max}^T values reported in Table 1, the topography-implied gravity contains signals at harmonic degrees much larger than N_{max}^T [*Hirt and Kuhn*, 2014]. To ensure spectral consistency among gravity values from both techniques, the additional spectral energy must be explicitly modeled in the SH approach (section 2.2.3), while it is implicitly accounted for in the NI (section 2.2.4).

2.2.2. Spatial Arrangement of Evaluation Points

In all cases, gravity is computed in terms of regularly spaced latitude ϕ /longitude λ grid nodes with 5 arc-min resolution (2160×4320 nodes). Regarding the radii r (vertical location) of the computation points, we use four different cases for each of the six N_{max}^T topography mass model, as follows:

- Case 1 Surface of the topography (Hsur). Evaluation points strictly reside at the surface of the topography $r = R_T$. About ~50% of points have a radius smaller than R , so are located inside the reference sphere.
- Case 2 Positive topographic heights (Hpos). Evaluation points reside at the surface of the topography only when outside R ; otherwise, evaluation takes place at R . About ~50% of points are therefore airborne, e.g., above deep depressions such as the Apollo basin, or over most of the lunar maria.
- Case 3 Brillouin sphere (H11km). We use a constant altitude of 11 km above $R = 1738$ km as safe-side approximation for the different Brillouin spheres radii R_B associated with the various N_{max}^T resolution levels and thus evaluate at $r = R_B = 1749 \text{ km} = R + 11 \text{ km}$. The chosen sphere safely encompasses all mass of the Moon. This also holds at high resolution, considering the highest peaks are ~10.8 km above R .
- Case 4 GRAIL sphere (H23km). As additional evaluation level, the average altitude of the GRAIL satellites during the extended mission phase of 23 km above $R = 1738$ km is used, so $r = R_G = R + 23 \text{ km}$. This places the evaluation points farther outside the field-generating masses than in case 3, giving some insights into strengths of the forward-modeled signals at GRAIL radius $R_G = 1761 \text{ km}$.

The four cases encompass scenarios where convergence may not be guaranteed for increasing spectral resolution (inside Brillouin sphere, cases 1 and 2) and those where convergence can be expected (outside Brillouin sphere, cases 3 and 4), e.g., *Hu and Jekeli* [2015].

2.2.3. Gravity From Spherical Harmonic Series

In the external SH approach, the topographic potential V is computed as series expansion of integer powers p of the topography [e.g., *Chao and Rubincam, 1989; Wieczorek, 2015*]

$$V_{nm}^{(p)} = \frac{4\pi R^3 \rho}{(2n+1)M} \prod_{i=1}^p (n-i+4) \frac{H_{nm}^{(p)}}{p!(n+3)} \quad (2)$$

$$V_{nm} = \sum_{p=1}^{p_{\max}} V_{nm}^{(p)} \quad (3)$$

where ρ is the adopted mass density value for the topography, M is the total mass of the Moon, R is the radius of the reference sphere, $p_{\max}$ is the maximum power considered in the series, $H_{nm}^{(p)} = (\overline{HC}_{nm}^{(p)}, \overline{HS}_{nm}^{(p)})$ denotes the SHCs of the function $(H/R)^p$ where $H = R_T - R$ are the topographic heights, $V_{nm} = (\overline{C}_{nm}, \overline{S}_{nm})$ are the potential SHCs evaluated to $p_{\max}$, and $V_{nm}^{(p)} = (\overline{C}_{nm}^{(p)}, \overline{S}_{nm}^{(p)})$ are the potential SHCs implied by the individual integer powers p ($1 \leq p \leq p_{\max}$) of the function $(H/R)^p$. The $V_{nm}^{(p)}$ can be thought of as contribution of the p th power of the topography to the total potential V_{nm} . The convergence of the SH series with parameter p is elaborated in detail in section 3.1 for the topographic potential models of various resolution levels.

The $(H/R)^p$ functions, computed in terms of Gauss-Legendre (GL) grids, are harmonically analyzed with the GL quadrature (GLQ), e.g., *Sneeuw [1994]*. Raising the function H/R that is band-limited from degree 0 to $N_{\max}^T$ to integer power p gives rise to additional short-scale signals with spectral energy in band of degree $N_{\max}^T + 1$ to $pN_{\max}^T$ [*Hirt and Kuhn, 2014*].

For each power p , the grid resolution dx (constant in longitudinal direction but variable in latitude direction, cf. *Sneeuw [1994]*) is chosen such that full recovery of these additional harmonics is possible: $dx = 180^\circ / (pN_{\max}^T)$. For instance, $dx = 2^\circ$ (91×181 nodes) is used for $N_{\max}^T = 90$, $p = 1$, while $dx = 30'$ for $N_{\max}^T = 90$ and $p = 4$. This is required not only to generate a largely complete SH model of the topographic potential [cf. *Hirt and Kuhn, 2014*] but also to avoid aliases that would come into play if the grid resolution dx was chosen smaller than $180^\circ / (pN_{\max}^T)$.

The maximum grid resolution used in the SHA is $1'$, corresponding to degree 10,800. The GLQ algorithm used is based on the SHTools package [*Wieczorek, 2015*] and software extensions as described in *Rexer and Hirt [2015]*. Their extension of the GLQ algorithm utilizes the *Fukushima [2012]* routines for stable computation of ALFs to ultra-high-degree, along with parallel processing capability.

Gravity values δg (defined here as radial derivatives of the potential V) are computed via

$$\delta g = \frac{GM}{r^2} \sum_{n=0}^{N^G} (n+1) \left(\frac{R}{r}\right)^n \sum_{m=0}^n (\overline{C}_{nm} \cos m\lambda + \overline{S}_{nm} \sin m\lambda) \overline{P}_{nm}(\sin \varphi) \quad (4)$$

where GM is the product of universal gravitational constant $G \approx 6.67384 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ and $M \approx 7.346304 \times 10^{22} \text{ kg}$, r is the radius of evaluation (as defined in section 2.2.2), N^G is the maximum harmonic degree in the synthesis of gravity values, and $N_{\max}^G$ is the maximum degree of the V_{nm} potential coefficients ($N^G \leq N_{\max}^G$).

When the radius of evaluation is constant (cases 3 and 4), software based on the GrafLab package [*Bucha and Janák, 2013*] was used, while for the cases with varying radii r across the lunar surface (cases 1 and 2), a modification of the isGrafLab software [*Bucha and Janák, 2014*] was used that deploys a Taylor series approach [*Hirt, 2012*] for the efficient computation of δg at the irregular lunar topographic surface. Both the GrafLab and isGrafLab software utilize the *Fukushima [2012]* routines for stable syntheses to ultra-high degree (e.g., 10,800).

2.2.4. Gravity From Numerical Integration

The NI technique evaluates Newton's integral [e.g., *Blakeley, 1996; Tenzer, 2005*] in the spatial domain. In this approach, the complexly shaped topography is discretized through an ensemble of regularly shaped mass elements (e.g., prisms and tesseroids, cf. *Heck and Seitz [2007]*, or polyhedra, cf. *Tsouliis [2012]*), and analytical expressions are used to accurately forward-model the gravity field of the mass elements. The gravitational effect of the complete topographic mass model is obtained through addition (superposition) of the gravity effects implied by all mass elements.

To describe the geometry of the input topographic mass model, topographic radii (equation (1)) are synthesized in terms of regularly spaced latitude-longitude grids with a very high spatial resolution of 30" (21,600 × 43,200 nodes). With the chosen resolution, the topography signal is massively oversampled, e.g., by a factor of 240 ($N_{\max}^T = 90$), and still, a reasonable oversampling of factor 10 is achieved for $N_{\max}^T = 2160$. The oversampling improves the spatial-domain representation of the topography signal (e.g., at least 240 nodes to represent a half wave-cycle when $N_{\max}^T = 90$), much reducing the impact of discretization errors on the computed gravity values. Different to the spectral technique (section 2.2.3), there is no need to explicitly model short-scale gravity signals (beyond the resolution of the input topography model $N_{\max}^T$) because these are "inherently" delivered by the NI approach.

For practical evaluation we use the highest grid resolution in the vicinity of each computation point and lower the grid resolution for remote masses. At each field (=computation) point, the numerical integration uses 30 arc-sec grid resolution within a 2° radius, 1 arc-min resolution within a 5° radius, 3 arc-min resolution within a 15° radius, and a 15 arc-min resolution for all grid points within 15 to 180° spherical distance to the computation point. This results in the evaluation of about 1.25 M mass elements for each computation point and ensures that the lunar topographic masses are completely considered in our (full-scale) numerical integration. The use of lower resolutions at some distance from the computation point is permissible, considering the quadratic attenuation of gravity with distance [e.g., Forsberg, 1984]. Importantly, the selected resolutions result in approximation errors that are at most at the microGal level when applied to the extreme case of a spherical shell with a thickness of 10 km.

We deployed Curtin University's in-house Newton integrator (as described in, e.g., Kuhn and Hirt [2016]) that uses a combination of prisms (near the computation point) and tesserooids (for distant masses), along with the concept of residual spherical shell integration [Kuhn et al., 2009] to limit the vertical extension of mass elements near the computation points. For the NI, advanced computational resources (made available by Western Australia's iVEC supercomputing facility) were utilized and a total of ~150,000 CPU hours used. Gravity values were computed at 5' resolution from each of the six topographic mass distributions (Table 1) at the four different levels of evaluation heights defined in section 2.2.2. Importantly, the numerical values of the constants (G, R, ρ) used in the NI are the same as in the SH modeling, ensuring consistency among both methods.

3. Numerical Study

3.1. Spectral Analyses

This section provides insight into the spectral constituents and characteristics of the six SH topographic potential models (Table 1). To investigate the spectral characteristics, we use dimensionless potential degree variances [e.g., Rapp, 1989]

$$\sigma_n^2 = \sum_{m=0}^n \left(\bar{C}_{nm}^2 + \bar{S}_{nm}^2 \right) \quad (5)$$

which are a measure of the spectral energy of the forward-modeled potential at a given harmonic degree n . Accordingly, degree variances are computed for the individual potential contributions $\bar{C}_{nm}^{(p)}, \bar{S}_{nm}^{(p)}$ with equation (5). Given there is no dependency on the radii of the field points, all potential coefficients from equations (2) and (3) and their power spectra from equation (5) refer to the reference sphere with the adopted reference radius $R = 1738,000$ m.

For each of the six models, Figure 1 displays the degree variances σ_n^2 for the individual contributions $V_{nm}^{(p)}$ of the integer power p (various colors) to the (total) topographic potential V_{nm} (black lines). For instance, Figure 1a shows that potential implied by the $N_{\max}^T = 90$ topography was computed through summation of the contributions made by the first 15 integer powers (equation (3)) up to a maximum degree $N_{\max}^G = 900$. According to the contribution scheme described in Hirt and Kuhn [2014], power $p=1$ contributes to $N_{\max}^T = 90$, power $p=2$ up to $2N_{\max}^T = 180$, and $p=15$ to degree 1350. Any spectral power at degrees $>N_{\max}^T$ originates from raising the topography to integer powers $p \geq 2$, and it is these "extra" short-scale energies that must be explicitly modeled in the SH approach if consistency with NI-based forward modeling is sought.

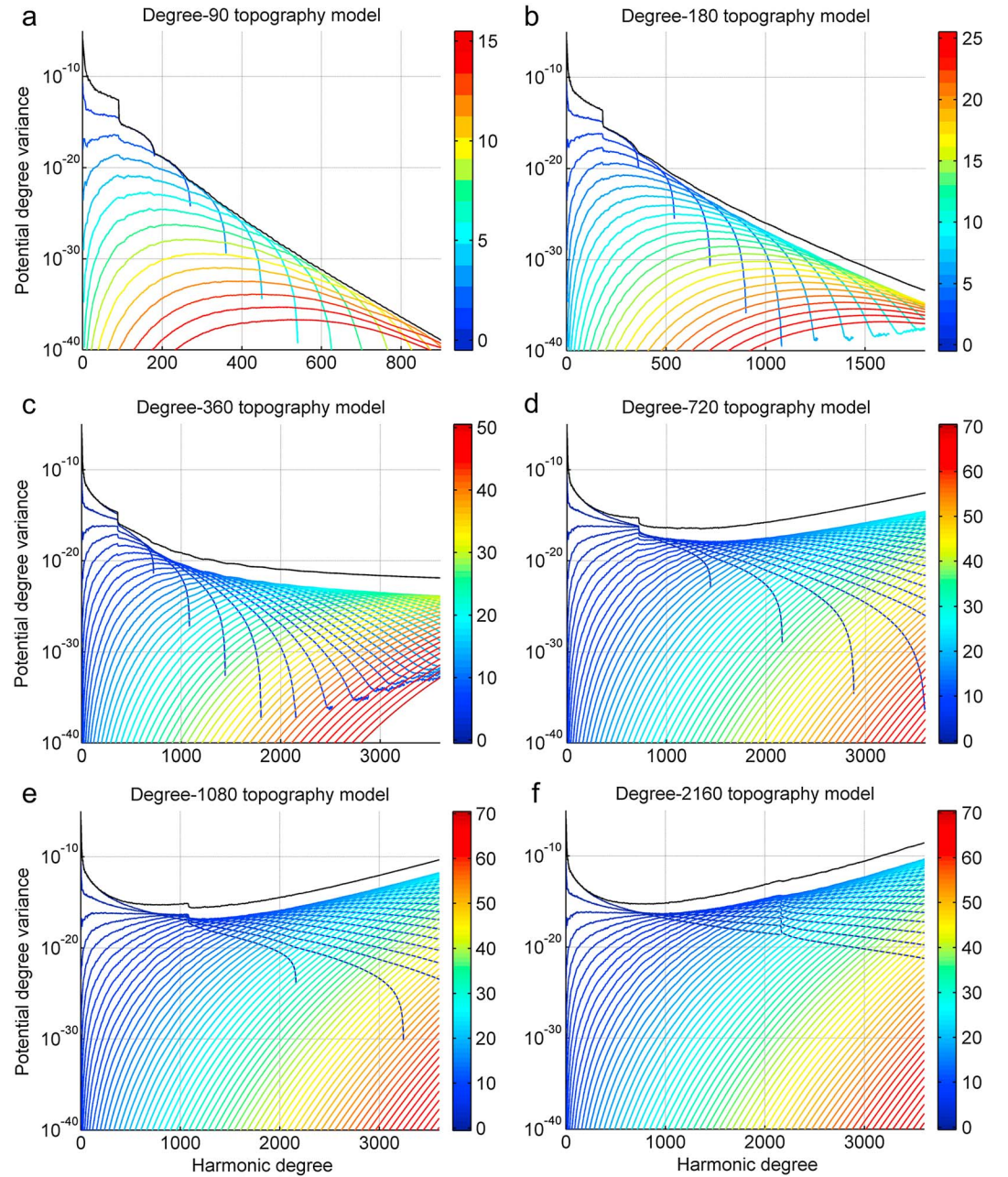


Figure 1. Contributions to the topographic potential made by the integer powers p of the topography (various colors) and (total) topographic potential (black), obtained as sum of the single contributions. (a) degree-90 input topography, (b) degree-180, (c) degree 360, (d) degree-720, (e) degree-1080, and (f) degree-2160 input topography. All panels show dimensionless potential degree variances (cf. equation (5)). All spectra refer to the reference sphere with $R = 1,738,000$ m.

For a comparatively low $N_{\max}^T$ value of 90, Figure 1a supports the postulation made in Wieczorek [2015] that the contribution made by “each succeeding term is smaller than the previous”. Figure 1a reveals for the degree-90 model fast convergence of the series expansion (equation (3)) with increasing p . For instance, the potential contribution of $p = 5$ does not exceed a degree variance $\sigma_n^2 = 10^{-20}$, and for $p > 10$ the spectral power is negligibly small ($\sigma_n^2 < 10^{-30}$), compared to the gravity signal associated with the linear term ($p = 1$) that contributes the bulk of spectral energy ($\sigma_n^2 > 10^{-15}$) to V_{nm} . Assuming that potential contributions with an associated signal strength $\sigma_n^2 < 10^{-25}$ are rather insignificant for the forward modeling [e.g., Hirt and Kuhn, 2012], Figure 1a shows that series convergence is reached for $p_{\max} \approx 7$ and $N_{\max}^G \approx 400$, with a monotonic decline of σ_n^2 with harmonic degree n visible over the whole spectrum.

When the spectral resolution $N_{\max}^T$ of the input topography model is increased, the convergence behavior of the SH series (equations (2) and (3)) changes considerably. For instance the computation of the implied potential from a $N_{\max}^T = 180$ model requires $p_{\max} \approx 12$ and $N_{\max}^G \approx 900$ to converge (Figure 1b). Further doubling to $N_{\max}^T = 360$ (Figure 1c) results in a much slowed-down convergence, with several integer powers to $p_{\max} \approx 40$ required to converge over the bandwidth considered (here $N_{\max}^G = 3600$, corresponding to the first 10 multiples of the input bandwidth $N_{\max}^T$). The spectral energy of the potential generated by $N_{\max}^T = 360$ is seen to steadily but very slowly decline beyond degree 360, with an energy level of $\sigma_n^2 = 10^{-20}$ reached near $n \approx 1300$ and $\sigma_n^2 = 10^{-22}$ near $n \approx 3600$ (black curve in Figure 1c).

From $N_{\max}^T = 360$ (Figure 1c) to $N_{\max}^T = 720$ (Figure 1d), the spectral characteristics of the potential models change substantially. Instead of monotonically decreasing energy, the potential generated by the $N_{\max}^T = 720$ topography shows a decrease to $n \approx 1300$, followed by a monotonic *increase* in spectral energy for degrees $n > \sim 1300$ (cf. Figure 1d). For the potential spectra of the $N_{\max}^T = 1080$ (2160) topography models, the increase in spectral energy enters at lower degree ($n \approx 800$) and is even more pronounced (Figures 1e and 1f). Given that gravity signal strengths are generally expected to decay with resolution [e.g., Rapp, 1989], this behavior is surprising. It may foreshadow divergence associated with the short-scale constituents of the potential model in the spatial domain inside the Brillouin sphere (section 3.2). As another explanation for this observation, the spectral-domain technique (equations (2) and (3)) extremely amplifies the short-scale potential constituents (at the chosen reference sphere with R the spectra in Figure 1 refer to), such that they attenuate “correctly” when gravity signals are computed outside the Brillouin sphere. Evidence supporting both hypotheses are obtained from our spatial-domain analyses in section 3.2.

A detailed look at the spectral energy associated with the various integer powers of the topography shows that for $N_{\max}^T \geq 360$, the largest potential contribution at short scales (e.g., $n \approx 3600$) is being made by higher order powers, e.g., $p \approx 35$ when $N_{\max}^T = 360$ and $p \approx 25$ when $N_{\max}^T = 2160$ (cf. Figures 1c and 1f) and the degree variances of V_{nm} may exceed those of any of the single contributions $V_{nm}^{(p)}$ by 2 orders of magnitude at high harmonic degrees (e.g., $n \approx 2000$), which is a result of the summation of several individual p contributions of similar spectral power (compare black versus color lines in Figures 1c to 1f).

Figure 1 also suggests that at any given degree n , the V_{nm} coefficients are spectrally completely modeled when the degree variance associated with the $p_{\max}$ -th contribution is at least 7 to 8 orders of magnitude below the maximum contribution. Using this criterion for the convergence of the summation over powers p (equation (2)), the V_{nm} coefficients are completely modeled for any $0 \leq n \leq 3600$, when $p_{\max} \geq 50$. Note that coefficients beyond $n > 3600$ are not further taken into account in our work because (i) they would possibly require very large integer powers p that might be swamped by under/overflow errors in a double-precision computation environment (M. Rexer, personal communication 2016) and (ii) they do not seem to influence the series behavior in a general sense.

As a side note, Figure 1 suggests that postulations made in Wiczeorek [2015] that “each succeeding term is smaller than the previous” and similarly in Wiczeorek and Phillips [1998] that “the magnitude of each successive term [...] is smaller than the previous one” hold only up to some certain harmonic degree. When the potential models reach or exceed a certain level of detail, the energy of several succeeding terms is larger than of the previous at high degrees as is seen from the spectral composition of the higher degree models (Figures 1c to 1f). As a result, there is an (apparent) increase in short-scale energy visible in the power spectra. We emphasize that all power spectra shown in Figure 1 refer to the reference sphere, so are not representative for the spectrum of, e.g., gravity values evaluated outside the Brillouin sphere, that would be declining with increasing harmonic degree. We note that in the context of ultra-high resolution Earth gravity modeling, the study by Balmino *et al.* [2012] made similar observations regarding the contributions of the higher order powers to Earth’s topographic potential.

Figure 2 finally shows the spectra of the six topographic potential models, as computed with equation (3) together with the degree variances of the GRAIL gravity model GRGM900C [Lemoine *et al.*, 2014]. Given (i) our forward-modeling constants, R, G, M (section 2.2) were adopted based on the constants of the GRGM900C model; (ii) the average mass density of the Moon’s crust well approximated by our adopted density value (cf. section 2.1); and (iii) the majority of the GRAIL-measured gravity signals is well explained through gravity from

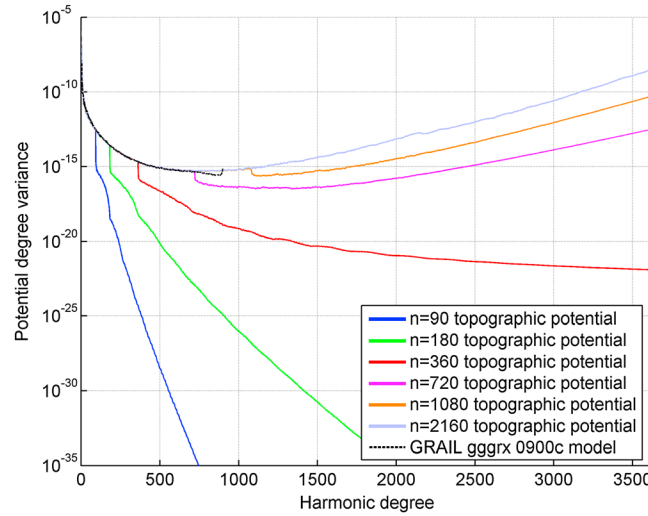


Figure 2. Comparison of potential degree variances of GRAIL GGGRX 0900c (model of the observed gravitational potential) with the topographic potential implied by degree-90 to degree-2160 topography models developed in this work.

$N_{\max}^T = 360$, and for larger $N_{\max}^T$, there is an apparent increase in spectral energy observed with increasing n . Additionally, the tails tend to be the more upward-curved, the larger $N_{\max}^T$, and this behavior might be an indicator for severe divergence of the series in the spatial domain (section 3.3).

3.2. Spatial Analyses

Gravity values δg from both forward modeling techniques, the spherical harmonic series (section 2.2.3) and the discretized numerical integration (section 2.2.4) were compared (a) as a function of the input topography resolution $N_{\max}^T$ (Table 1), (b) as a function of the maximum harmonic degree N^G in the gravity syntheses (equation (4)), and (c) for the four different vertical arrangements of computation points (cases 1–4, section 2.2.2). The descriptive statistics of the gravity values δg from both techniques, computed at 5 arc-min resolution (2160×4320 grid nodes covering the whole of the Moon), are reported in Table 2, and the statistics of differences $\Delta \delta g$ between gravity from both techniques are given in Table 3.

Maximum absolute differences, defined as $\max(|\Delta \delta g|)$, and root-mean-square (RMS) values of $\Delta \delta g$ are shown in Figure 3 for a total of 140 permutations over the three parameters $N_{\max}^T$, N^G , and evaluation heights. From top to bottom, the panels are arranged in terms of the increasing evaluation heights (Hsur, Hpos, H11km, and H23km, as defined in section 2.2.2), and each panel shows the convergence/divergence behavior for the six topography resolutions $N_{\max}^T$ (different colors) as a function of N^G (maximum degree in the gravity syntheses). In most cases, the N^G were chosen such that $N_{\max}^T \leq N^G = kN_{\max}^T \leq N_{\max}^G$, where $kN_{\max}^T$ are integer multiples of $N_{\max}^T$. The black solid line in all panels of Figure 3 marks the 1 mGal threshold that indicates sufficiently accurate representation of gravity through spectral gravity modeling with external harmonics.

As the key result of our work, Figure 3 gives unprecedented detailed insight into the divergence versus convergence behavior of the SH series of the Moon's topographic potential, exemplified here via gravity disturbances (cf. equation (4)).

Note on the terminology convergence and divergence as used in this study.

Here we refer to the term convergence when the residuals between numerical integration and SH series expansions (illustrated in Figure 3) show a steady decrease with increasing maximum spherical harmonic degree. Conversely, we refer to divergence when the residuals start to grow for increasing spherical harmonic degrees. We acknowledge that this interpretation does not relate to convergence or divergence of a series in a purely mathematical sense; it rather indicates the ability of the external spherical harmonic series (equation (4)) to accurately represent gravity in the space domain. In this case, the residuals fall below a given threshold (e.g., 1 mGal in Figure 3). On the other hand, technique limitations are clearly indicated by residuals

topography [e.g., Zuber *et al.*, 2013], the spectra of the forward models and GRAIL model are—as would be expected—in reasonably close agreement (Figure 2). See e.g., Lemoine *et al.* [2014] for a more detailed analysis of differences between observed and modeled gravity.

Importantly, Figure 2 reveals for all six topographic potential models the spectral power beyond the (nominal) resolution $N_{\max}^T$ of the input topography, that is, in spectral band $[N_{\max}^T + 1 \dots N_{\max}^G]$. For the six levels of resolution, these “tails” can be thought of as a set of curves with systematic upward-bending behavior (Figure 2). While for lower $N_{\max}^T$ the tails quickly drop (showing a rapid decay in short-scale energy), there is a slow decay for

Table 2. Descriptive Statistics of the Gravity Disturbances from NI and SH Modeling as a Function of the Location of Evaluation Points and Resolution of the Topography Model $N_{\max}^T$ ^a

Case Description		Newtonian Integration			Spectral Modeling With Multiples			
Location	Input $N_{\max}^T$	Min	Max	RMS	Output $N_{\max}^G$	Min	Max	RMS
Hsur	90	−1072.3	893.1	332.4	630	−1072.3	893.1	332.4
	180	−1179.7	917.0	339.2	1260	−1179.7	917.0	339.2
	360	−1190.1	1000.5	343.2	3600	−4.6E + 04	4.1E + 04	390.9
	720	−1173.3	1025.0	345.1	3600	−1.1E + 09	1.4E + 09	6.6E + 06
	1080	−1244.6	1032.4	345.8	3600	−4.3E + 11	7.1E + 11	1.2E + 09
	2160	−1283.5	1024.4	346.3	3600	−2.6E + 13	1.1E + 13	2.7E + 10
Hpos	90	−948.5	893.1	328.3	630	−948.5	893.1	328.3
	180	−1003.6	917.0	332.5	1260	−1003.6	917.0	332.5
	360	−972.2	1000.5	334.4	3600	−972.2	1000.5	334.4
	720	−976.4	1025.0	334.8	3600	−3.0E + 04	3.2E + 04	384.5
	1080	−976.2	1032.4	334.8	3600	−5.5E + 05	5.4E + 05	2.4E + 03
	2160	−976.3	1024.4	334.8	3600	−5.1E + 06	5.2E + 6	2.3E + 04
H11km	90	−859.4	866.3	317.2	630	−859.4	866.3	317.2
	180	−881.4	878.3	318.2	1260	−881.4	878.3	318.2
	360	−874.1	958.5	318.4	3600	−874.1	958.5	318.4
	720	−874.6	985.6	318.4	3600	−874.6	985.7	318.4
	1080	−874.6	991.0	318.4	3600	−874.6	991.0	318.4
	2160	−874.6	983.8	318.4	3600	−874.6	983.8	318.4
H23km	90	−796.1	770.1	306.9	630	−796.1	770.1	306.9
	180	−806.6	755.0	307.1	1260	−806.6	755.0	307.1
	360	−806.5	766.9	307.1	3600	−806.5	766.9	307.1
	720	−806.5	768.2	307.1	3600	−806.5	768.2	307.1
	1080	−806.5	768.2	307.1	3600	−806.6	768.2	307.1
	2160	−806.5	768.2	307.1	3600	−806.6	768.2	307.1

^aAll statistics Based on 5 arc-min global grids, unit is mGal in all cases.**Table 3.** Descriptive Statistics of the Gravity Differences “SH Minus NI,” as a Function of the Location of Evaluation Points, the Resolution of the Topography Model $N_{\max}^T$, and of the Resolution of the SH Modeling $N_{\max}^G$ ^a

Case Description		Truncated Spectral Modeling				Spectral Modeling With Multiples			
Location	Input $N_{\max}^T$	Output $N_{\max}^G$	Min	Max	RMS	Output $N_{\max}^G$	Min	Max	RMS
Hsur	90	90	−46.535	31.277	2.6425	630	−0.007	0.006	0.0006
	180	180	−79.455	54.772	4.8322	1260	−0.028	0.022	0.0016
	360	360	−130.168	91.242	6.0685	3600	−4.5E + 04	4.2E + 04	2.0E + 02
	720	720	−309.528	254.367	9.4200	3600	−1.1E + 09	1.4E + 09	7.0E + 06
	1080	1080	−3.1E + 03	2.6E + 03	38.197	3600	−4.3E + 11	7.1E + 11	1.2E + 09
	2160	2160	−2.6E + 07	3.3E + 07	1.2E + 05	3600	−2.6E + 13	1.1E + 13	2.7E + 10
Hpos	90	90	−34.652	31.277	2.3980	630	−0.027	0.037	0.0006
	180	180	−70.373	54.772	4.0004	1260	−0.072	0.188	0.0019
	360	360	−108.713	71.849	4.4135	3600	−0.655	0.724	0.0059
	720	720	−218.709	115.775	4.6511	3600	−3.0E + 04	3.3E + 04	2.0E + 02
	1080	1080	−401.508	343.723	5.8322	3600	−5.4E + 05	5.4E + 05	2.4E + 03
	2160	2160	−1.7E + 04	1.8E + 04	1.1E + 02	3600	−5.1E + 06	5.2E + 06	2.3E + 04
H11km	90	90	−20.984	15.220	1.2658	630	−0.014	0.004	0.0007
	180	180	−23.645	13.976	1.1787	1260	−0.018	0.013	0.0015
	360	360	−20.977	13.179	0.4664	3600	−0.030	0.031	0.0029
	720	720	−9.291	8.975	0.0918	3600	−0.068	0.042	0.0048
	1080	1080	−5.507	5.338	0.0254	3600	−0.145	0.181	0.0063
	2160	2160	−0.750	1.071	0.0075	3600	−0.172	0.260	0.0074
H23km	90	90	−9.159	6.688	0.6024	630	−0.019	0.005	0.0008
	180	180	−4.716	3.241	0.2860	1260	−0.021	0.005	0.0014
	360	360	−0.993	0.685	0.0313	3600	−0.024	0.005	0.0026
	720	720	−0.041	0.043	0.0045	3600	−0.029	0.005	0.0044
	1080	1080	−0.031	0.011	0.0053	3600	−0.031	0.011	0.0053
	2160	2160	−0.034	0.010	0.0063	3600	−0.034	0.011	0.0063

^aAll statistics based on 5 arc-min global grids, unit is mGal in all cases.

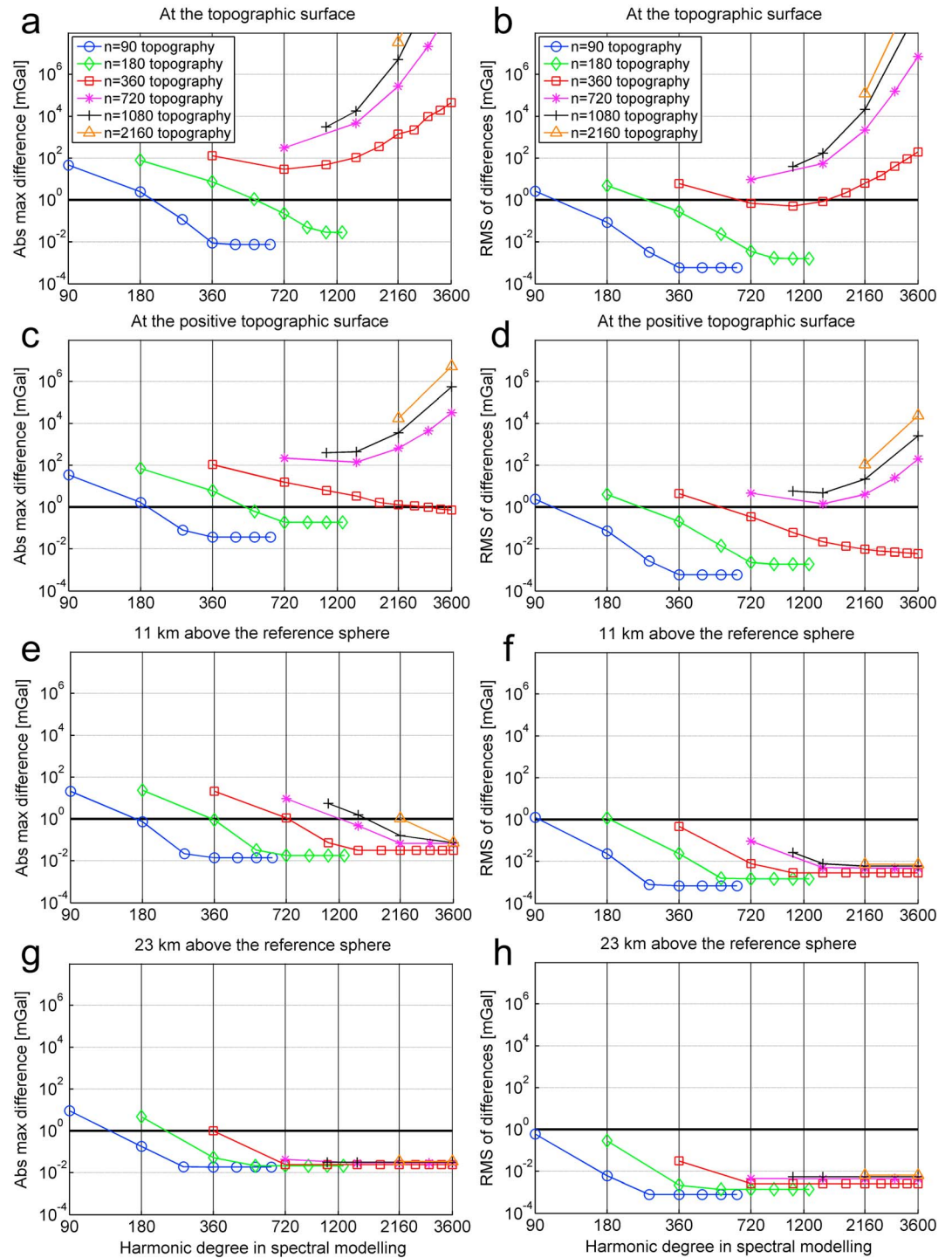


Figure 3. Comparison of gravity from SH and NI as a function of (i) the evaluation height (from top to bottom), of (ii) the resolution of the input topography model (various colors), and of (iii) the maximum SH degree considered in the spectral modeling. (left column) Absolute maximum of the differences SH minus NI and (right column) RMS (root-mean-square) of the differences. Comparisons based on 5 arc-min global grids; unit on the vertical axis is mGal. The figure shows how convergence and divergence depend on evaluation height and spectral resolution.

increasing with harmonic degree or failure to reach a given threshold (Figure 3), showing the inability of the external harmonic series expansion approach to adequately model near-surface gravity. While the indicators may point toward convergence or divergences, we cannot—in a rigorous mathematical sense—guarantee

series convergence or divergence when extending the harmonic series to infinity. This is because we work with truncated series expansions (that is, no evaluation takes place beyond degree $N_{\max}^G$ in equations (2) and (4)), as is a usual restriction when working with spherical harmonics.

3.2.1. Degree-90 and Degree-180 Topographies

For our lowest-resolution topography model ($N_{\max}^T = 90$), Figure 4a shows the gravity values δg from the numerical integration at 2160×4320 grid points across the Moon and Figure 4b the gravity differences $\Delta \delta g$ between the two techniques, whereby the SH series (equation (4)) was truncated to $N^G = N_{\max}^T$, and all computation points reside at the topographic surface with $N_{\max}^T = 90$. Figure 4b shows the spectral inconsistency between gravity from NI and SH when the latter is deliberately truncated to $N_{\max}^T$ in the synthesis. It is the short-scale signal produced by the higher order powers ($p \geq 2$) of the topography $H_{nm}^{(p)}$ at harmonic degrees $N^G > N_{\max}^T$ that is reflected by the differences in Figure 4b. In the present case, these short-scale signals may exceed ~ 10 mGal amplitudes (cf. Figure 4b and Table 3, “truncated spectral modeling”). When the spectral modeling is extended to $N^G = 7N_{\max}^T = 630$, the short-scale signals are fully modeled and the agreement between both techniques dramatically improves by ~ 4 orders of magnitude to the level of ~ 1 μ Gal (Figure 4c and Table 3, “spectral modeling with multiples”).

Figure 4c reveals excellent agreement that we interpret here as full convergence of the SH series anywhere across the surface of the Moon. The differences in Figure 4c reflect residual errors in the numerical integration which—if it was necessary—could be further reduced by increasing the oversampling (section 2.2.4). However, these residuals are entirely uncritical for this study, given their very small μ Gal-level amplitudes. Importantly, the excellent agreement of gravity values computed with two independent techniques from the same topographic mass distribution demonstrates the correct and consistent application of the two techniques and serves as a baseline (“best-case scenario”) for all further comparisons (Table 3).

Figure 4c also reveals that—probably in contrast to common belief—SH models can be accurately evaluated even deep inside the reference sphere (about 50% of the lunar surface is inside the reference sphere of radius R), provided that the resolution of the field-generating mass model is low enough. For points inside the reference sphere, that is, $r < R$, the factor $(R/r)^n$ —commonly referred to as attenuation or dampening factor in equation (4)—reverses its purpose and acts instead as amplification factor. If the gravity signals were not correctly amplified for evaluation points inside R , the agreement between the two methods would not have been as good as shown in Figure 4c. The duality of the amplification versus attenuation mechanism is not very widely addressed in the literature.

Table 3 reports the descriptive statistics for the comparisons of gravity from our $N_{\max}^T = 90$ and $N_{\max}^T = 180$ topography models at the four different altitudes, and Figure 3 shows the detailed convergence behavior of the spherical harmonic solution against the integration solution as a function of N^G (blue line for $N_{\max}^T = 90$, green line for $N_{\max}^T = 180$). In all cases—irrespective of the point location inside or outside the Brillouin sphere—the spherical harmonic series fully and accurately converge, which is seen from the $\Delta \delta g$ values being always smaller than 40 μ Gal (200 μ Gal for degree-180) for our complete spectral models (Table 3, “Spectral modeling with multiples”). Thus, our experiments suggest that the spherical harmonic gravity models converge anywhere outside the lunar topographic masses when the topography resolution is limited to scales of ~ 30 km ($N_{\max}^T \leq 180$).

3.2.2. Degree-360 and Degree-720 Topographies

Compared to the degree-90 and 180 comparisons, the increase to degree-360 resolution marks the “turning point” where not only the speed of convergence is reduced but divergence starts to become relevant. This is seen from the slowly decreasing discrepancies in Figure 3b, which do not drop below the ~ 0.7 mGal level when the $N_{\max}^T = 360$ model is evaluated to $N^G = N_{\max}^G = 3600$ at the topography but not inside R (case Hpos, Table 3).

When the evaluation points reside at the topography (case Hsur), the maximum absolute difference does not fall below ~ 70 mGal (see the local minimum in the red curve at $N^G = 720$ in Figure 3a) and rises with increasing N^G beyond the level of 10^4 mGal (Figure 3). We interpret this behavior as a clear sign of divergence of the spherical harmonic series attributable to the multiples of the input bandwidth.

The spatial distribution of evaluation points subject to divergence is displayed in Figure 5. Divergence primarily occurs over the deep Apollo basin, where the surface topography points are located deep (up to ~ 17 km) inside the Brillouin sphere (Figures 5a and 5c). Raising these deep evaluation points to R (points become

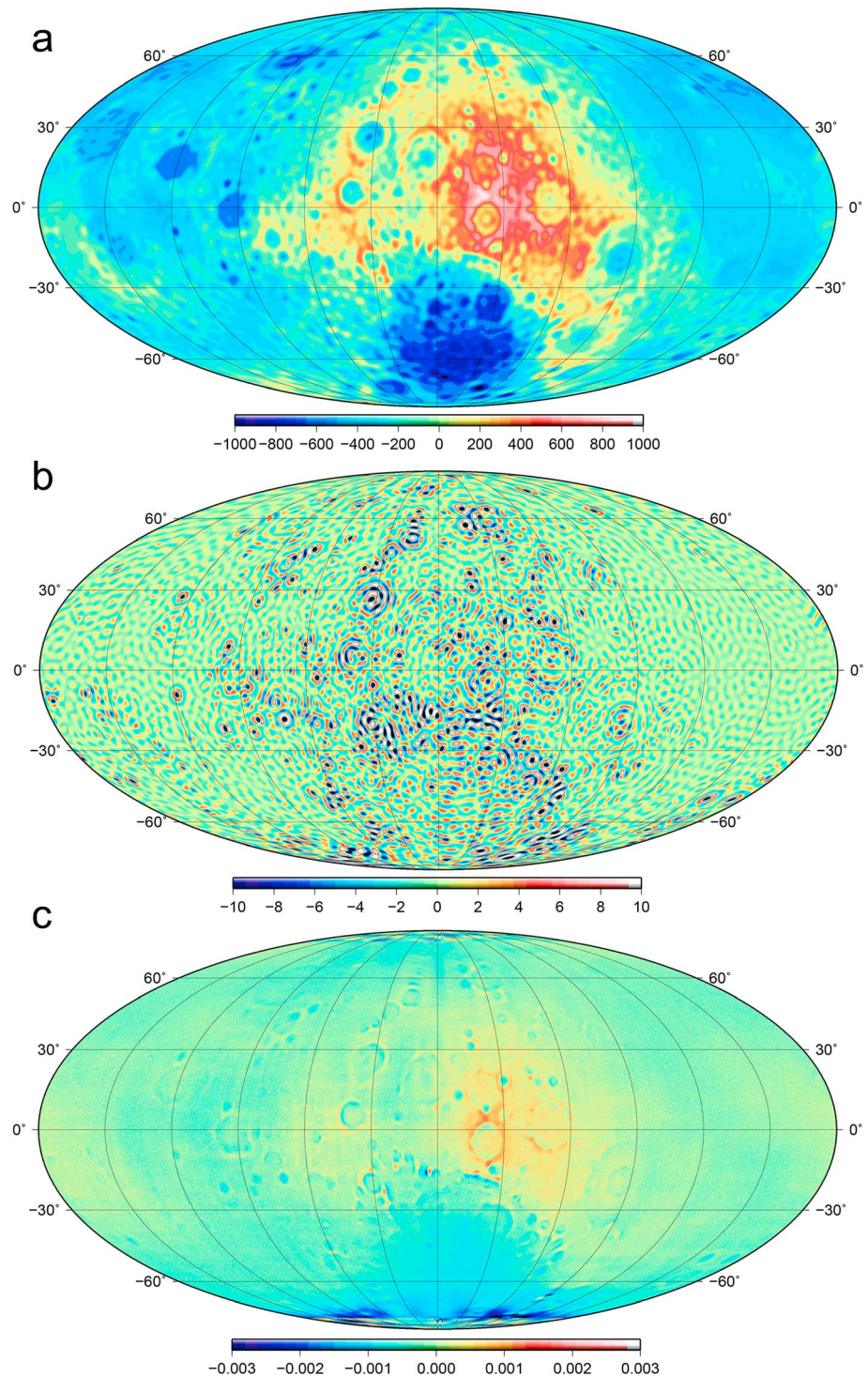


Figure 4. Forward modeling results using the degree-90 topography model. (a) Gravity disturbances from NI, (b) gravity differences “SH modelling to degree 90 minus NI,” and (c) gravity differences “SH modelling to degree 630 minus NI.” All gravity values computed at the topographic surface Hsur. (c) Importance of modeling short-scale gravity signals, beyond the resolution of the input model, to achieve excellent agreement between both techniques at the microgal level. Units in mGal.

airborne over, e.g., the Apollo basin) eliminates the divergence problem for the $N_{\max}^T = 360$ topography model (Figures 5b and 5d). This clearly illustrates the dependence of divergence on the depth of evaluation points inside R_B , and, more specifically, on $r < R$, where the $(R/r)^n$ factor amplifies the spectral energy.

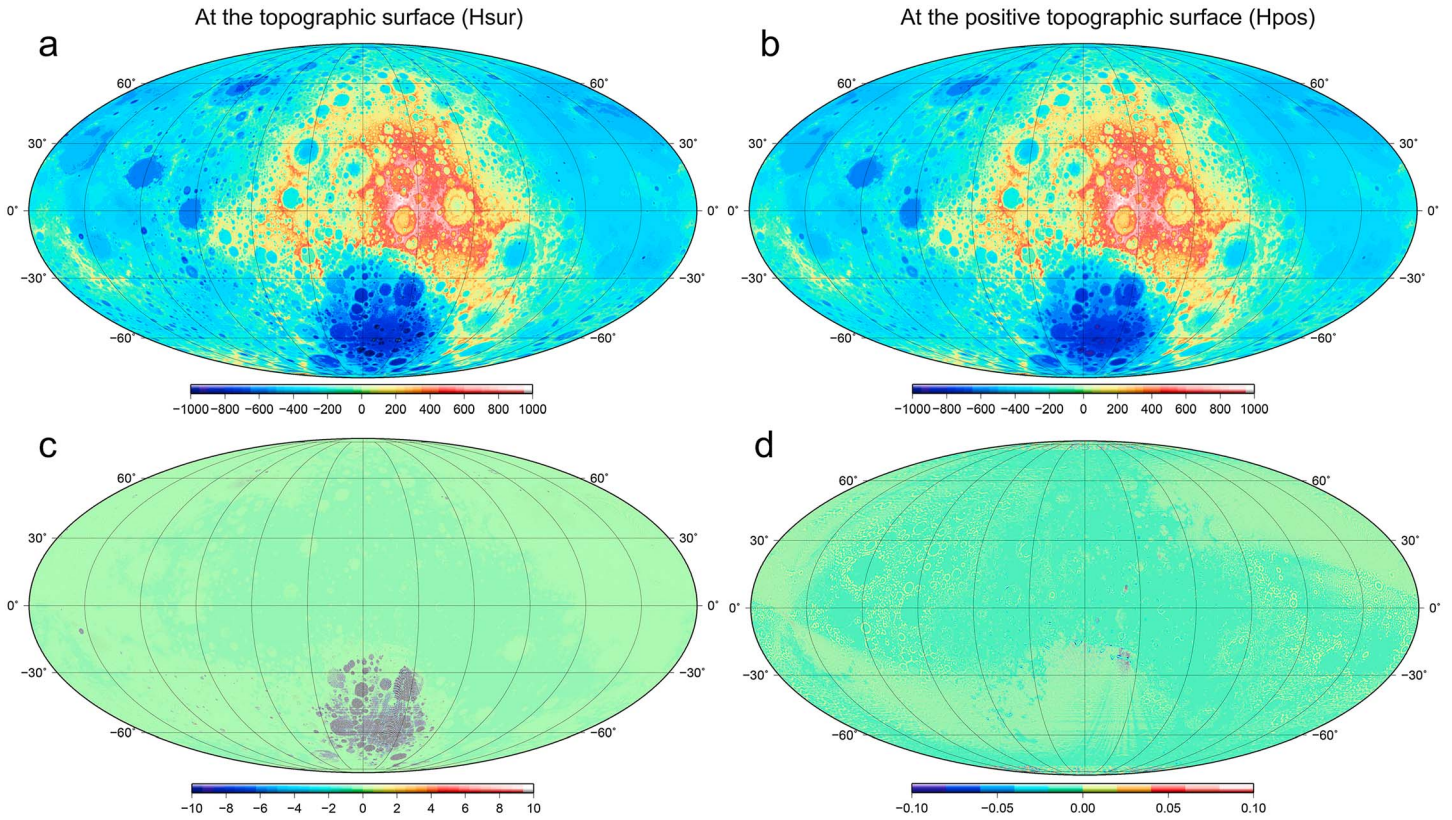


Figure 5. Forward modeling results using the degree-360 topography model. (a) Gravity from SH modeling computed at Hsur (positive and negative heights), (b) same but computed at Hpos (no negative heights), (c) gravity differences “SH minus NI” at Hsur, and (d) same but at Hpos. Maximum degree in the SH modeling is 2520. (left column) Divergence for computation points in the deep Apollo Basin, while (right column) convergence is reached when avoiding negative heights (i.e., lifting the computation points to 0 where $H < 0$). Units in mGal.

For our $N_{\max}^T = 720$ input topography model, the divergence in the spherical harmonic series is seen to exacerbate. For evaluation inside the Brillouin sphere (cases Hpos and Hsur), maximum absolute differences between gravity from both techniques always exceed 100 mGal (Figures 3a and 3c). A monotonic increase of the errors with the number of multiples considered in the gravity synthesis is attributable to divergence in the SH method (Figure 3a). For $N^G = N_{\max}^T$, the gravity differences reflect certainly (i) spectral inconsistency, and possibly (ii) emerging divergence, but discrimination between both effects is difficult.

For degrees of evaluation $N^G > N_{\max}^T = 720$, the discrepancies among both methods surge beyond 10^9 mGal (Table 3) and are present over most of the lunar surface (Figures 6a and 6c). For evaluation points at the topography, but not inside R , divergence effects are smaller but still exceed the 10^4 mGal level (Figures 6b and 6d). The comparisons between gravity from NI and SH reveal that divergence may originate over moderately undulating topography of the Moon’s nearside and “radiate” from what we call here a seed point over large parts of the planetary surface (Figure 6d). As a result, gravity values over large surface areas appear to become “contaminated” by divergence, the spatial patterns of which elude their prediction. As the central result of Figure 6, the SH technique is subject to severe divergence when applied to completely model the $N_{\max}^T = 720$ topographic gravity field at the surface of the Moon.

3.2.3. Degree-1080 and Degree-2160 Topographies

Gravity computed from the $N_{\max}^T = 1080$ (2160) topographies is severely affected by divergence when the spectral method is applied at computation points residing at the topography (Hsur). From Table 3, the maximum discrepancies exceed 3000 mGal ($N^G = N_{\max}^T = 1080$) and 3.3×10^7 mGal ($N^G = N_{\max}^T = 2160$). Modeling of degrees $N^G > N_{\max}^T$ worsens the disagreement with NI (see Table 3 and black/orange curves in Figures 3a and 3b).

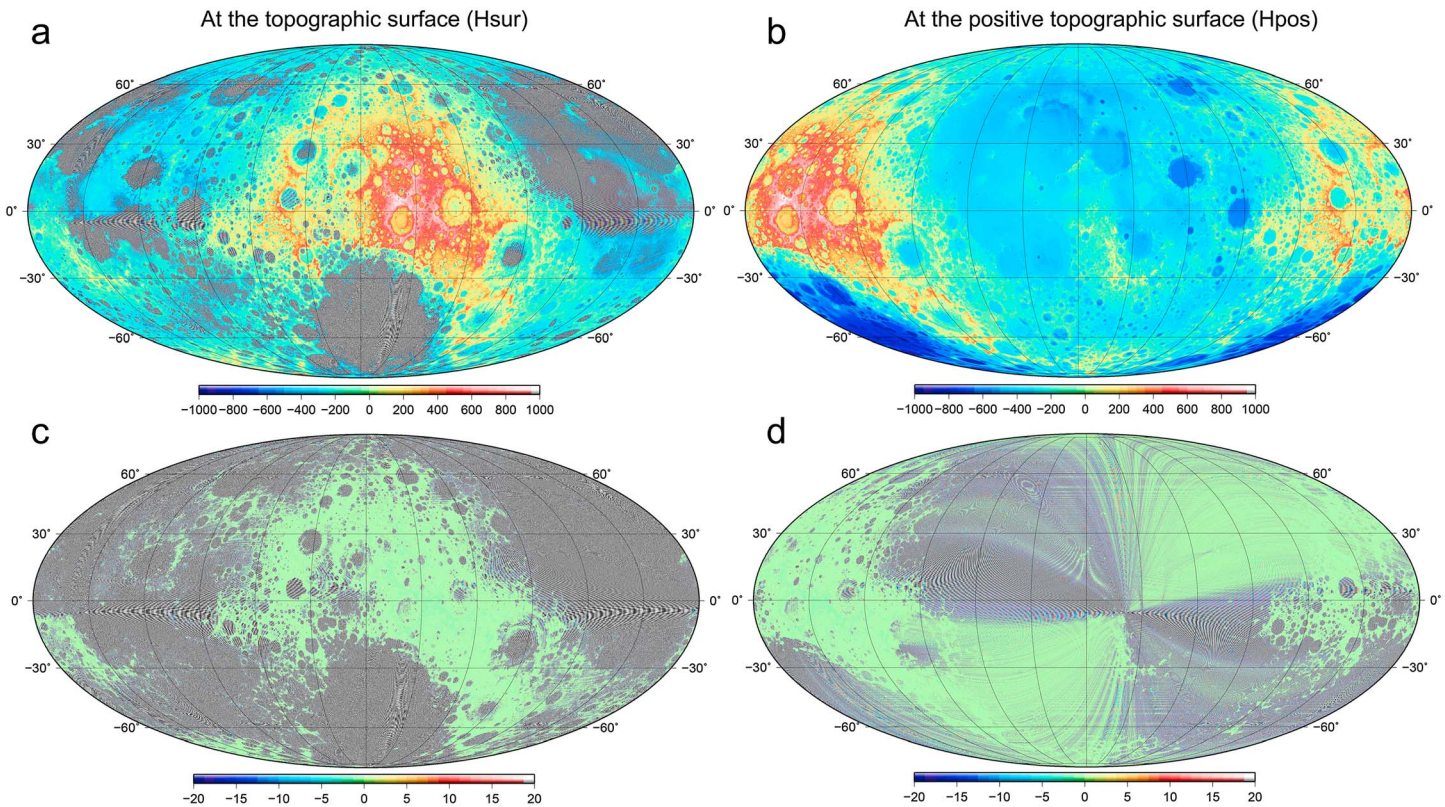


Figure 6. Forward modeling results using the degree-720 topography model. (a) Gravity from SH modeling computed at Hsur (positive and negative heights). (b) Same but at Hpos (no negative heights). (c) Gravity differences “SH minus NI” at Hsur. (d) Same but at Hpos. Maximum degree in the SH modeling is 2880. (left column) Severe divergence for most computation points at the topographic surface except at higher elevations. (right column) For Hpos, a seed point near the center of the nearside with divergence radiating over several 1000 km. The central meridian in Figures 6c and 6d has been chosen such that the spatial patterns of divergence become better visible. Units in mGal.

For $N_{\max}^T = 2160$ and computation points residing at the topography, Figure 7a shows gravity from NI and Figure 7b gravity differences between NI and SH modeling ($N^G = 2160$). From Figure 7b, the majority of computation points are severely affected by series divergence. Detailed views are given in Figure 8 over the far-side highlands, one of the—in a relative sense—lesser affected areas. Gravity from the spectral modeling is seen to provide grossly incorrect values inside craters (compare Figure 8a with Figure 8b). A characteristic feature of series divergence is extreme short-scale oscillations between large positive and negative values in craters that render the spectral technique inapplicable at such high spectral resolution over rough topography.

When the computation points are outside the Brillouin sphere (Figures 3e–3h), the agreement between the two gravity modeling techniques is better or much better than the mGal level, corroborating that both techniques deliver correct values that are accurate enough from a practical point of view. For $H = 11$ km and $N_{\max}^T = 1080$, the maximum difference is below 0.2 mGal when $N^G = 3600$ and below 0.3 mGal for $N_{\max}^T = 2160$ (cf. Table 3). The maximum differences could possibly be further reduced if the modeling was extended beyond degree 3600. However, this was not attempted in this work (see reasons in section 3.1).

3.2.4. Convergence Inside Versus Outside Brillouin Sphere

From top to bottom, Figure 3 gives detailed insight into the convergence behavior of spherical harmonic gravity models as a function of the evaluation height. For evaluation points inside the Brillouin sphere, our experiments (Figures 3a and 3c) demonstrate that the spherical harmonic series may converge or may diverge. Convergence is guaranteed for low spatial resolutions only (in our study up to $N_{\max}^T = 180$), while with higher resolution mass/gravity models, convergent series turn into divergent ones, and, as a result, invalid gravity values are obtained. The divergence effect is seen to be the stronger, the deeper the computation points reside inside the Brillouin sphere, and the finer the resolution of the gravity modeling (see Figures 3a and 3c).

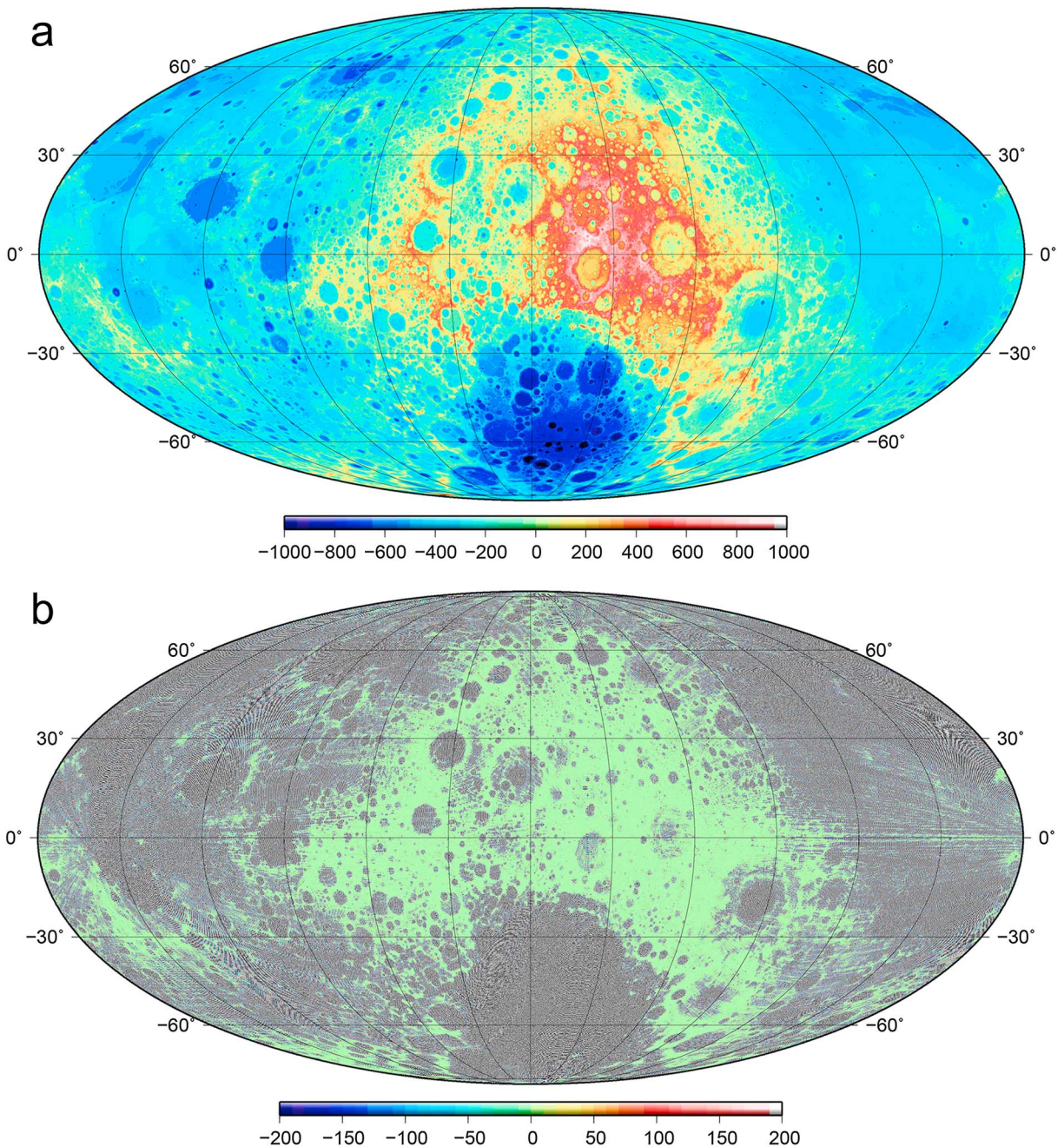


Figure 7. Forward modeling results using the degree-2160 topography model. (a) Gravity from NI. (b) Gravity from SH (evaluated to degree 2160) minus NI at the topographic surface, unit in mGal.

However, for evaluation points at R_B (Brillouin sphere, 11 km above R) or at GRAIL-altitude R_G (23 km above R), all SH series expansions are consistently seen to converge toward the NI solutions (Figures 3e and 3g). This is evident from the very good mutual agreement among the gravity values. The maximum absolute differences are always smaller than ~ 0.3 mGal (Table 3), and convergence for gravity from the $N_{\max}^T = 360$ topography is reached at degree $N^G = 1440$ at R_B and $N^G = 720$ at R_G . This is seen from the fairly horizontal graphs beyond the reported N^G values.

For gravity from the $N_{\max}^T = 720$ model, there is no need to evaluate gravity at R_B beyond $N^G = 2160$, while this value is lower at GRAIL altitude R_G ($N^G = 1440$) because the points are even farther away from the mass distribution, so short-scale signals are more attenuated as a consequence of Newton's law of gravitation.

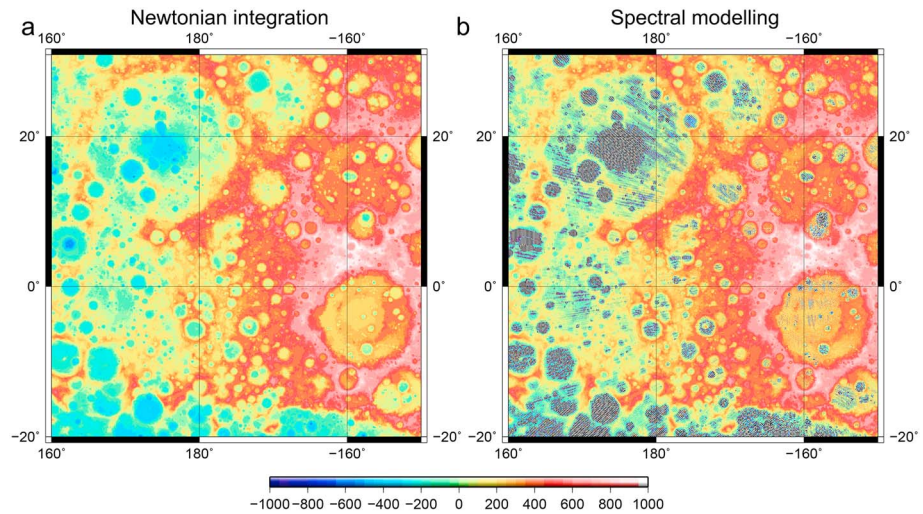


Figure 8. Forward modeling results using the degree-2160 topography model. (a) Gravity from NI. (b) Gravity from SH (evaluated to degree 2160) at the topographic surface, shown over the farside highlands, unit in mGal.

Thus, outside the Brillouin sphere, the series expansions converge against the independent NI solution in any case (Figures 3c and 3d), irrespective of the gravity model resolution or location of points outside that sphere. As such, the outcome of our experiments fully supports the theory of external spherical harmonics [Moritz, 1980; Takahashi and Scheeres, 2014; Hu and Jekeli, 2015].

3.3. The Relation Between Degree Variances and Divergence Inside the Brillouin Sphere—A New Hypothesis

Generally, convergence of exterior spherical harmonics inside the Brillouin sphere is rightly considered a highly “unstable phenomenon” [see Hu and Jekeli, 2015], and divergence behavior can be considered difficult to predict. However, a cross comparison between the degree variances of the topographic potential models (Figure 2) and the behavior of the spherical harmonic series expansions inside the Brillouin sphere (Figures 3a and 3b) suggests that the power spectra possibly contain valuable clues as to whether the series might be subject to divergence in free space bounded by the surface of the mass distribution and the Brillouin sphere. Relating Figure 2 with Figures 3a and 3c shows for topographic potential models of the Moon convergence anywhere at or outside the topographic surface if the spectral energy of the potential quickly decays with N^G (blue and green line, $N_{\max}^T = 90$ and 180), the possibility of divergence at points deep within the Brillouin sphere if the spectral energy shows very slow decay with harmonic degree N^G (red line, resolution $N_{\max}^T = 360$), and certainly occurring divergence somewhere inside the Brillouin sphere if the spectral energy associated with the input-bandwidth or multiples thereof rises for high harmonic degrees (e.g., purple line ($N_{\max}^T = 720$), beyond $N^G = 1200$; light blue line ($N_{\max}^T = 2160$), beyond $N^G = 800$).

Based on these observations, we formulate a new hypothesis to predict divergence of the spherical harmonic series, solely from the spectral-domain behavior:

A minimum in the degree variances of an external potential model foreshadows divergence of the spherical harmonic series expansions at points inside the Brillouin sphere.

This hypothesis is supported by all of our numerical experiments, so is valid for the gravity fields implied by our topographic mass models and the six levels of resolution investigated. However, for planetary gravity fields of bodies other than the Moon, further numerical tests are needed to in/validate our hypothesis.

One could be tempted to argue that the harmonic degrees N^G with ascending spectral power beyond the local minimum (Figure 2) should not be included in the gravity synthesis. However, for computation points outside the Brillouin sphere, our numerical study has demonstrated the validity and importance of—at least some of—these harmonics for the computation of correct gravity values. For instance, SH gravity from the $N_{\max}^T = 1080$ topography differs from NI by up to 5.6 mGal when evaluated to $N^G = 1080$ at the Brillouin sphere, while the maximum discrepancies are less than 0.2 mGal when $N^G = 3600$ (Table 3).

4. Discussion and Conclusions

The present study offers new detailed insights into the matter of divergence versus convergence of spherical harmonic series expansions of the gravitational potential, using forward-modeled gravity of Earth's Moon as example. A set of topographic mass models of varying resolution (from 61 to 2.5 km detail) was constructed, and the implied gravitational fields were computed at four sets of heights with spherical harmonic series and numerical integration, whereby the latter technique is not subject to the divergence problem. The numerical experiments considered the current state-of-the-art for technique comparisons (section 2), including very short-scale signal modeling in spherical harmonics and highly oversampled numerical integration to reach mutual consistency at the micro-Gal level. For all models and levels of resolution, spherical harmonic series of the gravitational potential were shown to be convergent outside the Brillouin sphere, as expected from the theory.

Our numerical study has shown the spherical harmonic series of the gravitational potential to fully converge in free space inside the Brillouin sphere to ~30 km resolution (degree 180) of the field-generating topographic mass model. Thus, gravity functionals from exterior gravitational potential models with similar resolution and characteristics can be adequately modeled with spherical harmonic series down to the lunar surface.

For mass models of degree 360 or higher resolution, divergence was shown to become relevant. For degree-360 models, the spherical harmonic diverge for evaluation points that are deepest inside the Brillouin sphere when the implied gravity field is computed with full detail; i.e., also, short-scale gravity signals beyond the degree-360 resolution are considered. For degree-720 resolution mass models, divergence prevents complete gravity modeling with spherical harmonics for the majority of surface topography points, and this exacerbates for degree-1080 and degree-2160 topography models used in the forward-modeling. The degree-2160 topography model divergence is evidently present when gravity is synthesized to degree-2160 at the lunar topography (Figures 7 and 8), preventing the use of the spherical harmonic technique—as described in section 2—from applications, such as prediction of gravity values at the lunar topography resolved to ~2.5 km. This finding is important, e.g., if degree-2160 spherical harmonic gravity models (with the high degrees predicted from topography) were to be used to compute gravity along spacecraft landing trajectories near the lunar surface.

For the degree-720 and degree-1080 resolution levels—which are commensurate to the resolution of GRAIL gravity field models—our study suggests that series convergence at the lunar topography should be considered doubtful. While our numerical experiments unambiguously reveal divergence occurring when the multiples of the input-bandwidth are evaluated, divergence cannot be excluded when the resolution of the implied gravity field model is truncated to those of the topography model.

Drawing an analogy between spectral gravity forward modeling for Earth and Moon using degree-2160 topography models, the dependency of series divergence on the ruggedness of topography becomes clear. For Earth, spherical harmonic series were shown to sufficiently converge at the surface of the topography (heights above zero) for degree-2160 topographic gravity models [Hirt *et al.*, 2016], while for the Moon, severe series divergence is evident for the gravity models of the same resolution class. This different behavior of the series for Earth and Moon needs further investigation in a follow-up study.

In order to exclude numerical accuracy problems as causes for the high-frequency behavior of the spectra shown in Figure 1 and the numerical results (Figures 3–8 and Tables 2 and 3), we have performed numerous tests for both the spectral domain approach (equations (2) to (4)) and the numerical integration technique. These included closed-loop tests for the former and comparison to analytical solutions for the gravitational effect of a spherical shell for the latter technique. Our tests conclusively showed that numerical problems cannot explain the results of this study. Furthermore, our results are in line with other studies on the divergence problem [e.g., Hu and Jekeli, 2015; Reimond and Baur, 2016] that attribute series divergence as cause for invalid values produced by the spectral technique inside the Brillouin sphere.

As a central result of this study, a new hypothesis was formulated that relates the spectral domain behavior of the gravity models to divergence in the spatial domain. Following our hypothesis, if the potential degree variances show a minimum, then the spherical harmonic series expansions diverge somewhere inside the Brillouin sphere. The hypothesis is fully supported by our numerical study for the Moon, but its generalization requires further modeling experiments for other planetary bodies, e.g., Mars and Earth.

In summary, by comparing the external potential series expansions (section 2.2.3) with numerical integration (section 2.2.4) for a set of topographic mass models with increasing level of detail (section 2.1), our numerical study (section 3) shows the cases in which the use of external potential series is good enough to accurately model the topographic potential and, conversely, the external potential series diverge, which is exactly when the internal potential cannot be neglected anymore in case the spectral modeling techniques are applied inside the Brillouin sphere.

The divergence is most likely the result of modeling the gravitational effect of masses above the computation point with external spherical harmonics, showing the importance of the internal potential in those cases.

Therefore, this paper has shown the limitations of the popular and widely used external potential expansions when used to model the near-surface gravity field implied by planetary topography beyond a certain level of resolution. To avoid divergence issues in such cases, either Newtonian integration (as done in our study) or the two-potential method (B. Bills, personal communication, 2017) can be used as alternatives. However, the combination of interior with exterior potential expansions, as done in the two-potential method, is radius-dependent (as noted in *Takahashi et al.* [2013, p 364]), potentially requiring several hundreds of coefficient sets to describe the gravity field implied by planetary topography. As such, both the two-potential method and Newtonian integration are computationally rather intensive and do not deliver solid spherical harmonic coefficients without further transformations (e.g., downward continuation of surface gravity). While our study showed the limitations of external harmonics near rough topography, it also indicates the level of resolution (in case of the Moon: $\sim 180^\circ$ or somewhat higher), when external potential modeling only is free of divergence, as measured by small differences to the Newtonian integration, such that the internal potential can be safely neglected. As future work, the two-potential method should be further explored and systematically compared with Newtonian integration.

5. Data statement

The models developed in this work will be made available via <http://ddfe.curtin.edu.au/models/> and also distributed via archives <http://icgem.gfz-potsdam.de/ICGEM/> and <http://geo.pds.nasa.gov/>.

Acknowledgments

This study was kindly supported by the German National Research Foundation through grant Hi 1760/01. We are grateful to Moritz Rexer who set up parts of the ultra-high degree software used in this study and to Blažej Bucha for discussions. We thank Bruce Bills for his constructive and detailed review and suggestion of the two-potential method and a second reviewer for his comments. Supercomputing resources were kindly provided by Western Australia's Pawsey Supercomputing Center.

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